

Homework 1

AMATH 582, Winter 2025

Assigned Jan 10, 2025.

Checkpoint Due on Jan 17, 2025. Report and Code Due on Jan 23, 2025 at midnight.

DIRECTIONS, REMINDERS AND POLICIES

Read these instructions carefully: There are two stages for submitting the assignment.

1. You will be submitting a *Checkpoint* approximately one week after the assignment was published (see due date above). The checkpoint includes submission of your Report and Code in progress (at least 1/3) (2 points) and taking the checkpoint quiz (3 possible points).
 2. You will be uploading a *Report (PDF)* to Canvas along with a zip of your *Code* (15 possible points). The grade will be based on how completely you addressed the problem as well as neatness and important things like: have you labeled your graphs and included figure captions (7 points will be given for the overall layout, correctness, and neatness of the report, and 8 additional points will be given for specific things that the TA will look for in the report and the code).
- The report should be a maximum of 6 pages long with references included. Minimum font size 10pts and margins of at least 1inch on A4 or standard letter size paper.
 - Do not include your code in the report. Simply create a zip file of your main scripts and functions, without figures or data sets included, and upload the zip file to Canvas.
 - Your report should be formatted as follows:
 - Title/author/abstract: Title, author/address lines, and short (100 words or less) abstract. This is not meant to be a separate title page.
 - Sec. 1. Introduction and Overview
 - Sec. 2. Theoretical Background
 - Sec. 3. Algorithm Implementation and Development
 - Sec. 4. Computational Results
 - Sec. 5. Summary and Conclusions
 - Acknowledgments (no more than four or five lines, also see the point below on collaborations)
 - References
 - L^AT_EX(Overleaf is a great option) is recommended to prepare your reports. A template is provided on Canvas in Homework/Files. You are also welcome to use Microsoft Word or any other software that correctly typesets mathematical equations and properly allows you to include figures.
 - Collaborations are encouraged; however, everything that is handed in (both your report and your code) should be your work. You are welcome to discuss your assignments with your peers and seek their advice but these should be clearly stated in the acknowledgments section of your reports. This also includes any significant help or suggestions from the TAs or any other faculty in the university. You don't need to give all the details of the help you received, just a sentence or two. A similar guideline applies to the use of Large Language Models (LLM). These are permitted for the study of topics and code presented in class and a better grasp of the problem and its solution. However, everything that is handed in (both your report and your code) should be your work and cannot be based on LLM content (modified or direct). Any use of external help should be specified in the acknowledgments section of the report.
 - **Late reports are subject to a 2 points/day penalty up to five days. They will be no longer accepted afterwards. For example, if your report is three days late and you managed to get 16/20, your final grade will be $16 - 6 = 10$, so be careful with late submission.**

PROBLEM DESCRIPTION: FINDING SUBMARINES

Your goal in this homework is to locate a submarine that is *moving* in the Puget Sound. We do not know much about this submarine as it is a new technology that emits an unknown acoustic frequency that you need to detect.

Broad spectrum recording of acoustics pressure data obtained over 24 hours in half-hour increments is available to you. You can download the data using the Google drive links on Canvas; either of the data files `subdata.npy` for Python users, `subdata.mat` for MATLAB users or `subdata.csv` in text format if the previous two formats are insufficient.

The data file contains a matrix with 49 columns of data corresponding to the measurements of acoustic pressure taken over 24 hours. These measurements are noisy (which is typically the case). The measurements themselves are 3D and taken on a uniform grid of size $64 \times 64 \times 64$. The provided (hwhelper) notebook will visualize this data for you and define the physical scales of the problem. *If the 3D plots are slow and you cannot see the dynamic behavior of the data consider downloading the GIF file `subdata.gif` on Canvas.*

SOME COMMENTS AND HINTS

Here are some useful comments and facts to guide you along the way.

1. First, observe that you are provided three-dimensional dynamic data, that is acoustic pressure measurements in 3D and as a function of time. This makes visualization difficult as the data set is effectively four-dimensional (3D+time). If you would like to see temporal variations then I suggest looking at slices of the data (eg isosurface command) as a function of time.
2. In class we only saw/ will see 1D and 2D Fourier transforms but here you may need a higher dimensional Fourier transform. Not much changes in the N-D setting except that you need to use the `fftn` function. `fftshift` remains valid and is still needed.
3. Recall Code Samples (FFT examples) and the discussion of the Gaussian function. We saw a useful fact about noise and Fourier transform that will help you in this assignment and in many applications:

It is known that adding mean zero white noise to a signal (Gaussian noise) is equivalent to adding mean zero white noise (Gaussian noise) to its Fourier series coefficients.

This fact enables one to devise a simple and effective "preliminary" noise filtering technique in situations where multiple measurements are available that are subject to the same noise. This is the case in imaging or acoustics applications like our submarine problem. Since the noise is random and mean zero it should average to zero over **many** samples. Thus, averaging the measurements in the Fourier domain is expected to reduce the noise.

The reduction will improve with an increasing number of aligned measurements, but in the case of our submarine, we only have a few measurements so you would still need to do additional filtering.

TASKS

Below is a list of tasks to complete in this homework and discuss in your report.

1. Through **averaging of the Fourier transform** determine the dominant frequency (center frequency) generated by the submarine. Verify your results through visualization.
2. Design and implement a **Filter** to extract this center frequency in order to denoise the data and determine a more robust path of the submarine. Visualize the denoised measurement the 3D path of the submarine and inspect the validity and effectiveness of the denoising.
3. Determine and **plot the** x, y coordinates of the submarine path during the 24 hour period. This information can be used to deploy a sub-tracking aircraft to keep an eye on your submarine in the future.